# *PROTEUS*: A Novel CRISPR-Cas-Based Cancer Cell Killer as a PDAC Therapeutic

## Introduction

KRAS-Driven Pancreatic Cancers as Effective Genetic-Level Cancer Therapeutic Targets

Pancreatic cancer is one of the most prevalent cancers worldwide and is estimated to be the second most common cause of cancer-related death by 2030 [1]. It has the highest mortality rate of all cancers worldwide, with an average 5-year survival rate of less than 12% after diagnosis [2]. Pancreatic cancer is difficult to detect in its early stages of development because the location of the organ makes it impossible to see or feel during routine medical examinations, and also since symptoms are often not observed until the tumor has already metastasized. Approximately 85% of tumors in the pancreas are non-resectable at the time of diagnosis; unfortunately, surgical resection is often the only feasible treatment despite its high probability of recurrence and the limited probability of long-term survival [3]. Today's therapeutic options are limited and unsuccessful in treating pancreatic cancer at an early stage, resulting in poor prognosis and increased dependence on surgical resection.

Over 90% of all detected cases of pancreatic cancer are pancreatic ductal adenocarcinoma (PDAC), an aggressive fatal malignancy due to its difficulty in early detection and the lack of viable treatment options [4]. PDAC falls under exocrine pancreatic cancer, specifically involving the mutation of exocrine cells along the lining of ducts in the pancreas [5]. Often, the exact cause behind the mutagenesis of these cells is unknown. Over 90% of known cases occur sporadically, owing to somatic mutations brought about by certain lifestyle habits. While not an exhaustive list, some risk factors include smoking, excessive alcohol consumption, diet, obesity, and diabetes [6]. In general, these risk factors act by introducing carcinogens (nitrosamines, polycyclic aromatic hydrocarbons, etc.) into the circulatory system, or induce health-related problems such as pancreatitis or pancreatic hypertrophy and hyperplasia, encouraging the onset and proliferation of pancreatic cancer driver mutations [7]. The remaining 10% of cases are attributed to genetic inheritance. Specific mutations in genes related to syndromes such as hereditary breast and ovarian cancer syndrome, Lynch Syndrome, hereditary pancreatitis, and cystic fibrosis, among many others, make up roughly 20% of these cases. These genetic predispositions may increase an individual's risk of developing PDAC from 2- to 132-fold compared to the general population [5].

Treatment for PDAC depends on the patient's general health and progression of the cancer. Among the standard treatment options, the most effective treatment for improving life expectancy is the surgical removal of the tumour [8]. Several surgery options are available including pancreaticoduodenectomy (the Whipple procedure), total and distal pancreatectomy. However, few patients are diagnosed early enough to benefit from surgery, and even fewer are eligible to receive this treatment [9]. Once the tumour has metastasized, palliative surgery is often done to minimize pain and improve the patient's quality of life [10]. Other standard treatment options include chemotherapy, a drug treatment that circulates throughout the body via the bloodstream, and radiation, a more localized therapy where intensive electromagnetic waves are directed towards tumour cells [8]. These methods (either standalone or combined) aim to cease the growth and proliferation of cancer cells, damaging the genes in their nucleus or disabling the biochemical mechanisms involved in cell division [11]. However, the effectiveness of these therapies becomes limited as the cancer advances and spreads to adjacent organs. At the cellular level, the dense stromal microenvironment of PDAC acts as a physical barrier against drug delivery agents, preventing the penetration of standard drug treatments such as gemcitabine into the tumour mass [4, 12]. Drug resistance also occurs at the genetic level, which can lead to relapse after a certain amount of time [13]. As a result, for many patients in the advanced stages of pancreatic cancer, these treatments are performed as a palliative measure.

As opposed to standard care treatments, which target hallmarks of cancer, gene therapies have shown great promise as a treatment modality by addressing the root cause of cancer. Early forms of gene therapy included gene transfer, in which modified genetic material was introduced into native cells either to fight the disease or to replace the altered disease-causing gene [14]. Since then, gene therapy has expanded to include immunotherapy, oncolytic virotherapy, and gene-editing technologies [15], introducing deficient, missing, or under-expressed therapeutic genes, or inhibiting the expression of deleterious genes [16]. In the case of the latter, one issue that arises is that deleterious driver mutations may be expressed in some cancer cells of the host, but not in others given cancer's heterogeneous nature. This often prevents genetic-level therapies from providing a complete treatment given the potential for relapse due to non-targeted cancer lines. In the case of PDAC, however, one gene tends to be mutated in nearly all manifestations of the disease; a key mediator of tumor formation in the pancreatic duct lining is a mutation of the Kirsten rat sarcoma virus oncogene homolog (KRAS), specifically the KRAS G12V mutation. PDAC is considered one of the most RAS addicted cancers with a near 100% KRAS mutation frequency [17]. The KRAS G12V mutation changes the second nucleotide of codon 12 from GGT to GTT, resulting in an amino acid change from glycine to valine. This mutation causes a conformational change of the GTP-binding site, consequently reducing the intrinsic rate of GTP hydrolysis [18]. Given the high mutation frequency, KRAS mutations in PDAC represent an effective target for genetic-level cancer therapeutics.

## CRISPR-Cas Cancer Therapeutic Strategies

Precision medicines provide many advantages over traditional cancer therapeutics in diminishing off-target effects, increasing treatment efficacy, and often avoiding cancer drug resistance [19, 20]. One promising treatment modality in development is CRISPR-Cas-based precision medicines. Employing a gRNA complementary to target loci in a patient's genome or transcriptome, CRISPR-Cas systems can be tailored to perform a variety of functions in response to detection of a certain nucleotide sequence. The specific response elicited depends on the CRISPR-Cas system employed in the therapeutic; native CRISPR-Cas systems are often re-engineered to optimize their functionality [21, 22]. CRISPR-Cas-based cancer therapeutics today have focused on interrupting or preventing cancer cell proliferation by cleaving, nicking, degrading, or manipulating oncogenic genes or transcripts, as is done with Cas9 and CAs13-based tools [21 22, 23]. Based on our review, however, no research has attempted to pair CRISPR-Cas genetic/transcriptomic detection with proteolytic activity to prevent cancer's development. Here, we take advantage of the novel gRAMP (giant Repeat Associated Mysterious Protein) CRISPR-Cas effector, which forms a complex with the Csx29 protease to cleave fusion proteins integrating Csx29's substrate, Csx30. These fusions, when cleaved, induce pyroptotic cell death similar to native gasdermins.

<u>gRAMP-Csx29 CRISPR-Cas Protease as a Potential Cancer Therapeutic</u>

The gRAMP effector is classified as the Type III-E CRISPR-Cas system. Type III CRISPR-Cas effectors are typically assembled from multiple protein subunits; this enables the overall effector to carry multiple functions, including gRNA binding, target RNA cleavage, DNA cleavage, and secondary messenger synthesis [24]. Type III-E is a recently identified atypical type III system. It encodes a large polypeptide (gRAMP) as a fusion of four Cas7-like domains, one Cas11-like domain, and a big insertion domain (BID), but lacks Cas10, the signature component of a canonical type III system that is required for secondary messenger production [24, 25]. Subsequent studies demonstrated that the gRAMP ribonucleoprotein (RNP) complex is capable of RNA-guided RNA cleavage at two specific sites that are six nucleotides apart [24, 26]. Unlike the type VI CRISPR-Cas effector Cas13, gRAMP does not cause collateral RNA cleavage and has no cytotoxicity in eukaryotic cells [24, 27], providing a togglable system with substantial potential as a precise, transcriptomic-level therapeutic.

Although the Type III-E effector employed here was originally derived from the *Desulfobacteraceae* bacterium *Desulfonema ishimotonii* [28, 29], we integrated the system into yeast and mammalian cell lines for *in vitro* validation of a prospective human therapeutic. gRAMP natively recognizes complementary viral transcripts, then activates ancillary immune proteins. gRAMP uses CRISPR RNA (crRNA) to guide target RNA recognition and cleaves single-stranded RNA [26]. It also physically combines with the caspase-like Csx29 peptidase (also called TPR-CHAT due to its Tetratricopeptide Repeat (TPR) domain and Caspase HetF Associated with TPRs (CHAT) domain) to form the gRAMP-TPR-CHAT complex, also known as Craspase (for CRISPR-guided caspase) [24]. The protease activity of Csx29 provides *Desulfonema ishimotonii* colonies viral immunity by inducing cell death of individuals infected by bacteriophage, similar to the abortive infection strategy of caspase-mediated cleavage of bacterial gasdermins [26, 30, 31, 32, 33]. Gasdermins generate an antiphage defense system in bacteria by inducing pyroptotic cell death [32]; after activation via caspase or caspase-like protein cleavage in a linker region between a lipophilic N-terminal domain and large inhibitory C-terminal domain, the gasdermin N-terminal domain oligomerizes to form membrane pores that release inflammatory cytokines and induce cell lysis [24, 32]. They bind specifically to acidic phospholipids such as phosphatidylinositol phosphates and phosphatidylserine, which localize on the inner leaflet of the plasma membrane [34]; this allows gasdermins to form pores in the host's lipid bilayer, but prevents targeting of neighbouring cells.

Eukaryotic cells encode structural and functional homologs of bacterial gasdermins, though there are approximately 50 bacterial gasdermins that form a clade distinct from that of eukaryotic homologs [32]. There is less diversity in the gasdermin families of eukaryotic species. Humans, for example, have six proteins in the family: Gasdermin A through F. Mice lack Gasdermin B, but have three isoforms of A and four of C [34]. Regardless, gasdermins represent a conserved, homologous family across all domains of life [35]; the N-terminal domains of mammalian and bacterial gasdermins exhibit conservation of a twisted central antiparallel β sheet with connecting helices and strands throughout the periphery [32]. The most divergent element is the interdomain linker region between the C-terminal autoinhibitory domain and the N-terminal pore-forming domain [35]. This is greatly due to the unique cutsite for each gasdermin, which differs based on the caspase involved in cleavage. In *Desulfonema ishimotonii*'s Craspase system, the Csx29 protease cleaves another ancillary protein, Csx30, between residues M427 and K428, separating a large N-terminal fragment of ~48-50 kDa from a small C-terminal fragment of ~15-16 kDa [28, 30]. While it has been demonstrated that cleavage of Csx30 triggers abortive infection by inducing cellular toxicity [36], the lack of homology with known proteins makes the mechanism of this induced toxicity, and thus the precise physiological function of Csx30, difficult to infer with confidence based on current literature [24]. Regardless, the caspase-like protein cleavage of Csx30 provides an opportunity to employ the Craspase system for modular proteolysis of gasdermins.

Using an optimally truncated Csx30 (trCsx30) from Strecker et al [28]., we developed chimeric proteins with human gasdermin genes, removing part of the intrinsically disordered linker region and inserting the trCsx30. We hypothesized that insertion of trCsx30 with the M427-K429 cutsite into human gasdermins would enable the Craspase system to cleave the linker region of human gasdermins upon recognition of its target RNA, inducing pyroptosis in the host cell. This system would provide a novel treatment modality for cancer gene therapy by targeting high-frequency driver mutations in conserved cancers such as KRAS G12V PDAC and inducing inflammatory cell death in cancer cells. Pyroptosis also releases inflammatory molecules that alert surrounding tissue of the presence of cancer; cytokines released during pyroptosis, including IL-1β and IL-18, promote lymphocyte recruitment, induce IFNγ production, activate T cells, macrophages, and natural killer cells, and enhance the immune system's ability to kill tumor cells [37, 38, 39]. Thus, this treatment would both eradicate cancer cells as well as immunologically prime the body through the inflammatory effect of pyroptosis. By directly triggering pyroptosis in solid tumours, this treatment offers a double-pronged approach to anti-tumor therapy and has the potential to be synergistically coupled with other immunotherapies such as checkpoint blockade, personalized cancer vaccines, and CAR T Cell therapy.

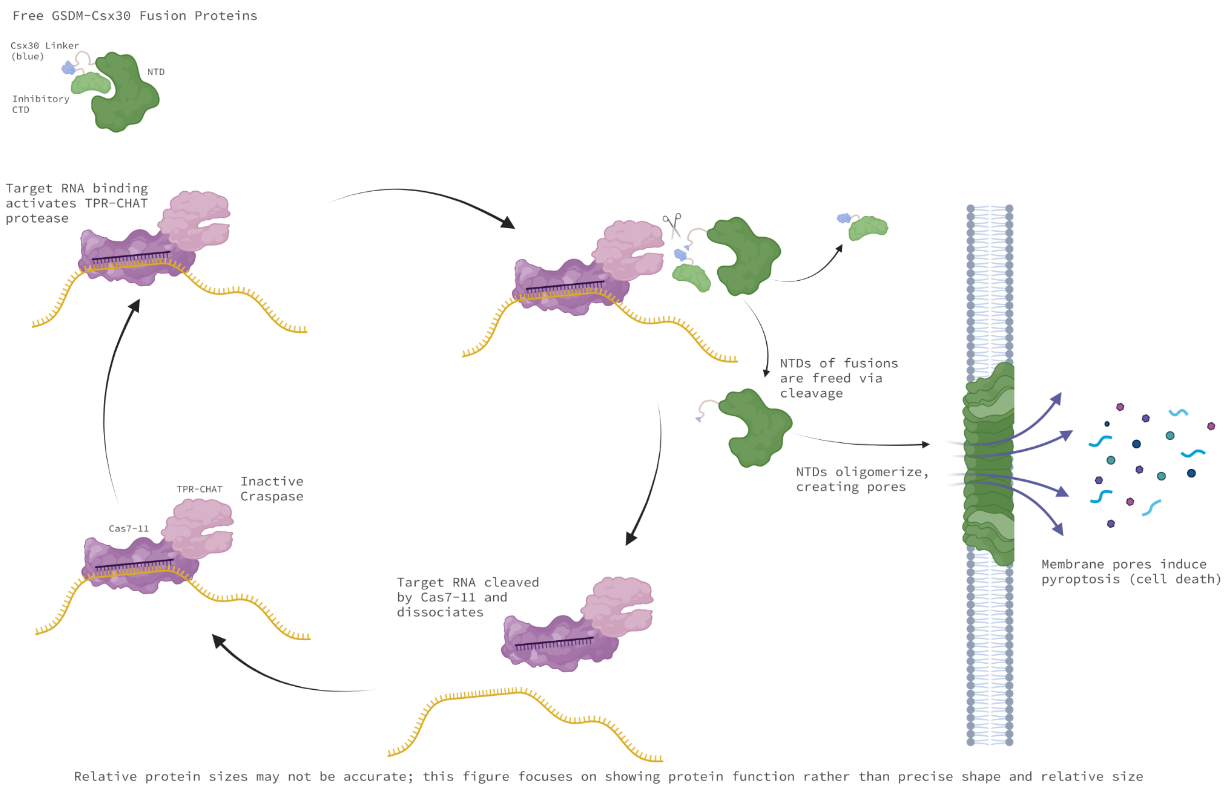



Figure 1: Craspase binding its target oncogenic mRNA, cleaving a gasdermin-Csx30 fusion protein, creating membrane pores in the cell which induce pyroptosis.

We developed several gasdermin-trCsx30 fusion proteins where trCsx30 was inserted into the disordered linkers of both human Gasdermin D (GSDMD) and E (GSDME). Fusion proteins involving GSDME are referred to as GDEx30 followed by an identification of the specific fusion protein, and GSDMD fusion proteins are similarly called GDDx30 followed by an identification of the specific fusion. The following fusion proteins were engineered and subsequently tested:

- GDEx30 N-terminal insertion: insertion of trCsx30 in the N-terminal side of the disordered linker, inserted between residues R249 and I250 of GSDME.
- GDEx30 C-terminal insertion: insertion of trCsx30 in the C-terminal side of the disordered linker, inserted between residues D279 and G280 of GSDME.
- GDEx30 30% replaced: replacing the center 30% of the total disordered linker (15% of each side of the center of the link) of GSDME with trCsx30 (from L255 to A264).
- GDEx30 CaspRem: replacing the site where GSDME is natively cleaved by caspases – D267 to A271 inclusive – with trCsx30.
- GDDx30 N-terminal insertion: insertion of trCsx30 in the N-terminal side of the disordered linker, inserted between residues P243 and A244 of GSDMD.
- GDDx30 C-terminal insertion: insertion of trCsx30 in the C-terminal side of the disordered linker, inserted between residues G281 and G282 of GSDMD.

- GDDx30 30% replaced: replacing the center 30% of the total disordered linker (15% of each side of the center of the link) of GSDMD with trCsx30 (from S252 to M265).
- GDDx30 CaspRem: replacing the site where GSDMD is cut by caspases – F272 to G276 inclusive – with trCsx30.

In all cases where the native gasdermin caspase cutsite was not eliminated in the construct, the cutsite was mutated via PCR to deactivate its potential for spontaneous cleavage by host caspases.

We also developed several vectors containing the KRAS wild type gene, KRAS G12V, and the Craspase machinery. For the Craspase system, we used a nuclease-dead variant of Cas7-11 in our experimentation. This was done primarily to prevent RNA cleavage; this allows for oncogenic transcripts to remain intact after detection by (and subsequent dissociation from) the Craspase system. Higher concentrations of the target transcript allow for sustained detection by individual Craspase units, increasing the efficacy and likelihood of a pyroptotic event. Another reason for using a dead mutant was to prevent Csx29 deactivation; Craspase exhibits self-regulatory behaviour through target RNA cleavage. After the system cleaves its target, it switches the Csx29 peptidase off, thereby timing the duration of the immune response and possibly recycling the Craspase complex to bind new target RNAs [24]. In order to create a functional cancer therapeutic using Craspase, it is necessary to have sustained, high efficacy targeting of oncogenic transcripts. By preventing RNA cleavage, Craspase is switched on indefinitely, or until it spontaneously dissociates from its target RNA. In other words, gRAMP is prevented from deactivating Csx29's proteolytic activity, enabling a single Csx29 to theoretically cleave an unlimited number of GSDM-Csx30 fusions while the Craspase is bound to its target sequence.

To validate this therapeutic *in vivo*, we sought to express each component of the system separately to test for toxicity to the host, then verify appropriate cleavage of the fusions upon Craspase recognition of its target. We assembled and tested 7 standardized vectors (with modular parts for facile re-engineering) for unprovoked cellular toxicity in both a yeast model – *Saccharomyces Cerevisiae* – as well as in mammalian cell lines – HKP1 mouse lung cancer cells, MiaPaca 2 and Capan-1 human pancreatic cancer cells. After verifying the innocuity of the components in the host cells, the full system was tested using the KRAS G12V mutation as a proof of concept. This involved transforming the host with gRAMP (expressed constitutively), Csx29 (galactose-induced), one of the eight fusion proteins (galactose-induced), and the KRAS G12V gene (cyanimide-induced). Each fusion protein was tested separately to determine which would grant the highest efficacy in terms of pyroptosis induction. Using scrambled and complementary guide RNAs (gRNAs) for the Craspase system, we determined the baseline background cleavage of Csx30. We also tested both wild type KRAS and KRAS G12V to determine the varying levels of cleavage with a single base pair (bp) mutation in the target. This would be essential knowledge down the line to ensure the safety of the system *in vivo*; if several bp mismatches are tolerated by the system, it would be necessary to take into consideration the potential near-complementary loci throughout the host's genome that could be targeted, and make the necessary adaptations to prevent off-target effects.

Successful induction of pyroptosis was measured in several fashions. For all experiments, we used growth curves to estimate cell viability with 30-minute increments over 24 hours. We used propidium iodide and SYTO 9 staining to label cells as living or dead. In tandem, we performed flow cytometry to categorize subgroups as well as further validate the results from growth curves. The results of our experiments provide promising *in vitro* validation of the system as a potential pancreatic cancer therapeutic targeting cancer at the transcriptomic level. Compared to existing methods, the Proteus system offers both preventive and curative treatment capability; those at high risk of developing pancreatic cancer may seek treatment in advance and those with early-stage pancreatic cancer may be treated prior to the cancer metastasizing. The inherent modularity of the system also opens the door for personalized cancer treatments; using next generation sequencing to characterize cancerous versus non-cancerous reads from a patient, gRNAs targeting the cancer while limiting off-target effects throughout the patient's genome could be developed and substituted into the system. Using a viral vector delivery method, the system could then be introduced into the patient, providing an effective means to eradicate the cancer in its early stages. The future steps of the investigation are to perform *in vivo* experimentation in syngeneic mouse models to elucidate the immunological consequences of the system on the host, as well as the potentially varying efficacies in-patient compared to *in vitro*. Potential limitations of future extensions include the possibility of inducing hyperinflammatory effects in the host from frequent pyroptotic events, the necessity for a vector with a substantial genetic payload given gRAMP's large size, and the potential for cancers to down-regulate expression of the system *in vivo*. Potential extensions of the work include employing the system as a preventative, or "genetic vaccine", against viral diseases such as AIDS, meningitis, and HTLV-induced adult T-cell lymphoma. Only after *in vivo* experimentation is completed will it become clear how promising this novel therapeutic modality will be.

## Materials and Methods

The plasmids and primers used in this investigation are listed in Table 1 below. Primers described in Table 1 were used for polymerase chain reaction (PCR) amplification using either Q5 High-Fidelity DNA Polymerase from New England Biolabs (NEB), or Platinum SuperFi DNA Polymerase from Thermo Fisher Scientific. DNA fragments were ordered from Integrated DNA Technologies (IDT) as gBlock Gene Fragments. Several DNA assembly strategies were used to clone the constructs, including Gibson assembly, Golden Gate assembly, and restriction digestion and ligation. Transformation was performed using NEB's High Efficiency Transformation Protocol for C2987H and C2987I 5-alpha Competent *E. Coli* strains before sequencing the constructs for validation through Genome Quebec. Validated constructs were then transformed into *S. Cerevisiae* for all experiments.

<u>Cloning: Construction of Vectors and Transformation into *E. Coli*</u>

Figure 2 below provides an overview of the finalized vectors developed in the workflow. Csx30 in pGal-MLs-SCLS (hereafter referred to as the pYes vector) serves as a control to confirm cleavage of the protein when KRAS G12V is transcribed. Wild type KRAS (KRAS wt) in pYes will be used to determine whether the KRAS G12V-targeting Craspase system will have tolerance for bp non-complementarity; KRAS wt and KRAS G12V differ by 1 bp. Thus, cleavage of KRAS wt by KRAS G12V-targeting gRNA suggests tolerance for mismatches between the gRNA and the target transcript. The mutant KRAS in pYes encodes KRAS G12V, which will be used to demonstrate that the presence of KRAS G12V mRNA activates Csx29's proteolytic activity. pML107 (Addgene plasmid #67639) encodes both Cas7-11 and the KRAS G12V gRNA, which will be essential for cleaving Csx30 and Csx30-GSDM fusions in the presence of KRAS G12V mRNA. Csx29 in pESC-HIS (hereafter referred to as the pEsc vector) encodes the protease that complexes with Cas7-11; without Csx29, Cas7-11 is unable to cleave Csx30 or Csx30-GSDM fusion proteins. Finally, both GSDMD and E under pYes are the vectors encoding the Csx30-GSDM fusions. Based on constructs inserting Csx30 into different regions of the linker of the GSDMs, we will determine the most efficient fusion protein for the system.

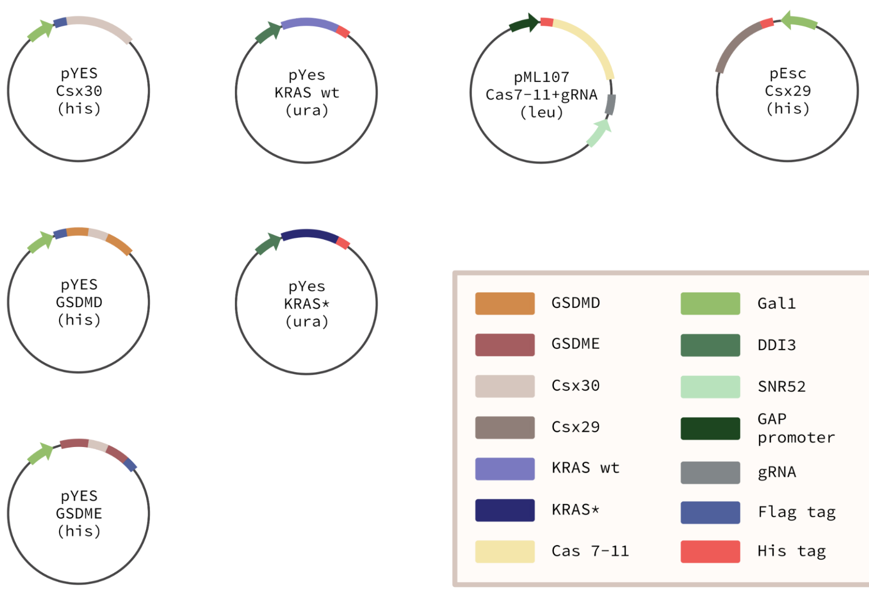



Figure 2: Finalized vector constructs for the system. Csx30 in pYes can be induced by the GAL1/GAL10 bidirectional yeast promoter. The HIS3 gene in the backbone serves as an auxotrophic selection marker. Csx30 is also tagged by a FLAG tag at its N-terminus for blotting once expressed *in vitro*. GSDMD and E fusion protein vectors are similarly in pYes with the HIS3 marker, all galactose-induced through GAL1/GAL10. GSDMD fusion vectors are FLAG tagged at the N-terminal end, while GSDME fusions are C-terminal FLAG tagged. KRAS wt in pYes and KRAS G12V in pYes are both His tagged, induced by the cyanamide DDI2/3 promoter. These vectors have the URA3 selection gene. Cas7-11 and the gRNA under pML107 are constitutively expressed by the GAP and SNR52 promoters respectively. The vectors include the LEU2 gene as its selection marker. Cas7-11 is His tagged at its N-terminal end. Csx29 under pEsc includes the GAL1/GAL10 promoter and the HIS3 selection marker. Csx29 is His tagged at its N-terminal end.

The starting vectors were pYes, pEsc, and pML107. pYes was used for developing the fusion proteins, the control Csx30 plasmid, and the KRAS vectors. Only the Csx29 protease vector was developed with pEsc. Finally, pML107 was used for the Cas7-11-gRNA vectors. In all cases, the Cas7-11 was maintained, while varying primers were employed to introduce unique gRNAs into pML107 via PCR amplification. We took advantage of oligo pools to test binding efficiency of a wide variety of gRNAs with both complete complementarity and partial complementarity to its target transcript. Csx30 was extracted from its source vector, pCDNA3_up1-3xHA. GSDMD and E were derived from pET-SUMO-hGSDMD and pLVX-Puro-hGSDME respectively. The KRAS G12V gene was ordered as a gBlock. Cas7-11 was isolated from pDF0234 pCMV - huDisCas7-11 dead mutant. Finally, the Csx29 protease gene was originally from the host plasmid pCDNA3_Csx29. A schematic of the cloning strategy to develop the finalized constructs is provided in Figure 3 on the following page:

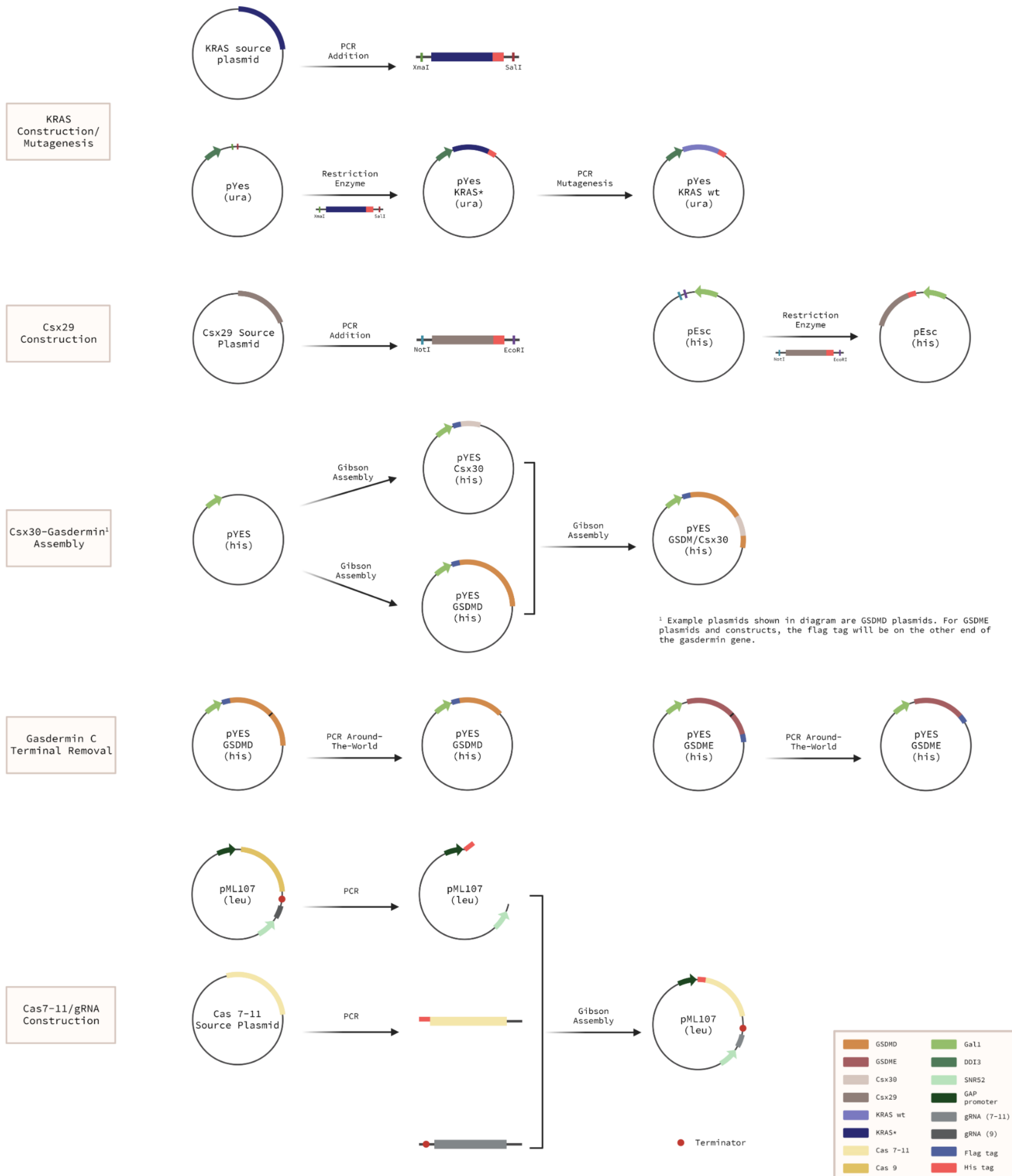



Figure 3: (KRAS Construction/Mutagenesis): a gBlock containing the cyanamide-induced DDI2/3 promoter and the mutant KRAS G12V gene was inserted into the pYes vector via Gibson assembly. To produce a separate pYes vector with

wild type KRAS, PCR mutagenesis was performed on the KRAS G12V pYes plasmid. (Csx29 Construction): the Csx29 gene from pCDNA3_Csx29 was amplified out of its vector via PCR, then restriction digested and ligated into the Invitrogen pCR2.1-TOPO vector, after which it was digested from pCR2.1-TOPO for ligation into pEsc. (Csx30-Gasdermin Assembly): Csx30 was inserted into an empty pYes backbone from its source vector, pCDNA3_up1-3xHA, via Gibson assembly. Gasdermin D and E genes were inserted in separate pYes backbones in parallel from their source plasmids, pET-SUMO-hGSDMD and pLVX-Puro-hGSDME respectively, via Gibson assembly. Through subsequent Gibson assemblies, the Csx30 gene was inserted into the gasdermin genes to create four gasdermin D constructs and four gasdermin E constructs. The four constructs for each involved inserting Csx30 into specific loci of the linker regions of each protein (at the N-terminal end, C-terminal end, caspase cutsite, and a construct replacing 30% of the linker region with Csx30). Several gasdermin-Csx30 constructs were assembled to enable determination of the highest efficiency locus for fusion protein cleavage in downstream experiments. (Cas7-11/gRNA Construction): The Cas7-11 gene from pDF0234 pCMV - huDisCas7-11 dead mutant was amplified out of its vector via PCR, adding sequences at each end of the gene to enable a downstream Gibson assembly. The same was done to remove the Cas9 gene from the pML107 vector (which would be replaced by the Cas7-11 gene). Finally, a 3-way Gibson assembly was performed with the linearized pML107, Cas7-11, and several scrambled and complementary gRNA gBlocks (where each gRNA would have its own construct).

A gBlock containing the cyanamide-induced DDI2/3 promoter and the mutant KRAS G12V gene was inserted into the pYes backbone with the URA3 uracil auxotrophic selection marker gene (rather than the TRP1 tryptophan marker gene from the original pYes vector) via PCR using primers KRAS_pYes_SynthInsert_F and KRAS_pYes_SynthInsert_R. The gBlock contained a yeast-optimized Kozak sequence, a C-terminal 6xHis-tag, and a stop codon along with the KRAS gene. In order to express wild type KRAS, the KRAS G12V in pYes was mutated through PCR mutagenesis using primers KRAS_D12G_mPCR_F and KRAS_D12G_mPCR_R. This PCR mutagenesis introduced a point mutation corresponding to conversion of Aspartic acid (D) to Glycine (G) at the 12th amino acid residue of K-Ras. Once the wild type and mutated constructs were effectively prepared, both pYes-KRAS plasmids were transformed into C2987H 5-alpha Competent *E. coli.* Following plating, colony PCR was performed on several proliferative colonies. The amplified constructs were then electrophoresed on an agarose gel. Bands of equivalent size to the expected construct were extracted then sent for DNA sequencing to validate the construct.

For transformation of Csx29, the Csx29 gene from plasmid pCDNA3_Csx29 developed in Hu et al. was isolated using primers Csx29_Nterm_mPCR_F and Csx29_CTerm_mPCR_R. These primers additionally introduced an initiator codon, a yeast-optimized Kozak sequence, an N-terminal 6xHis-tag, a stop codon, and EcoRI and NotI restriction sites for downstream applications. The extended Csx29 gene was then inserted via EcoRI and NotI restriction digestion into pCR2.1-TOPO from Invitrogen (hereafter referred to as pTOPO). As with the KRAS constructs, the pTOPO-Csx29 vector was transformed in C2987H 5-alpha Competent *E. coli*, plated, amplified through colony PCR, and electrophoresed on agarose gel. The DNA was extracted and the construct was verified via DNA sequencing. Csx29 was then digested from pTOPO (again using restriction enzymes EcoRI and NotI) for ligation into the pEsc plasmid, which includes the HIS3 histidine auxotrophic selection marker gene and the GAL1/GAL10 bidirectional yeast promoter. Again, the pEsc-Csx29 vector was transformed, plated, PCR-amplified, electrophoresed, the DNA was extracted, and the construct was sequence-verified.

For the Csx30-Gasdermin fusion proteins, Csx30 was inserted into the pYes backbone from its source vector, pCDNA3_up1-3xHA, using Gibson assembly. The primers used were Csx30_MCS_Csx30_GB_F, Csx30_MCS_Csx30_GB_R, Csx30_MCS_pYes_GB_F, and Csx30_MCS_pYes_GB_R. These primers additionally added a yeast-optimized Kozak sequence, a start codon, and an N-terminal FLAG-tag. In parallel, Gasdermin D was inserted into a separate pYes vector from its source vector, pET-SUMO-hGSDMD (Addgene plasmid #111559), via Gibson assembly. The primers used were GSDMD_MCS_GSD_GB_F, GSDMD_MCS_GSD_GB_R, GSDMD_MCS_pYes_GB_F, and GSDMD_MCS_pYes_GB_R. These primers added a yeast-optimized Kozak sequence, a start codon, and an N-terminal FLAG-tag. Additionally, Gasdermin E was inserted into another separate pYes vector from its source vector, pLVX-Puro-hGSDME (Addgene plasmid #154876), via Gibson assembly. The primers used were GSDME_MCS_GSD_GB_F, GSDME_MCS_GSD_GB_R, GSDME_MCS_pYes_GB_F, and GSDME_MCS_pYes_GB_R. These primers added a yeast-optimized Kozak sequence, a C-terminal FLAG-tag, and a stop codon. Two families of vectors were then created by subsequent Gibson assembly: a Gasdermin D-Csx30 vector family and a Gasdermin E-Csx30 family. This involved inserting Csx30 into a prespecified region of the linker in each of the Gasdermin proteins. Each family had four vectors: an N-terminal insert, where the Csx30 was inserted at the N-terminal end of the linker region without removal of any amino acids in the gasdermin; a Caspase Cutsite Replace insert, where the Csx30 was inserted precisely at the location where the native human caspase cleaves the given gasdermin, replacing the cutsite amino acid; a C-terminal insert, where the Csx30 was inserted at the C-terminal end of the linker region without removal of any amino acids in the gasdermin; and a 30% Replace insert, where the Csx30 was inserted after amino acid T251 in Gasdermin D and Y253 in Gasdermin E, then the subsequent 24 amino acids of D and 16 of E were replaced with the Csx30. The primers used for the Gasdermin D-Csx30 fusion vectors and Gasdermin E-Csx30 fusions can be found in Table 1 in Supplementary Information (from ID No. 17 to 48). All vectors included the HIS3 histidine auxotrophic selection marker gene and the GAL1/GAL10 bidirectional yeast promoter.

A novel construct was developed for co-expression of the catalytically dead Cas7-11 (dCas 7-11) and gRNA complementary to KRAS G12V mRNA. This construct was developed using the pML107 vector as a backbone, replacing the Cas9 gene with dCas7-11 and the Cas9 gRNA with Cas7-11 gRNA. Cas7-11 was PCR-amplified from its source plasmid, pDF0234 pCMV - huDisCas7-11 dead mutant, then inserted into the linearized, truncated pML107 backbone along with a gRNA gBlock. In order to introduce different gRNA spacers into the vector, we introduced BsaI sites in the gRNA to allow for Golden Gate assembly of new spacers. We ordered oligo pools from IDT with overhangs complementary to the sticky ends of the digested backbone; upon restriction digestion by BsaI, a new gRNA spacer can be introduced into the pML107 backbone using an equivalent protocol to the BbsI guide oligo insertion shown in Ran et al. [40].

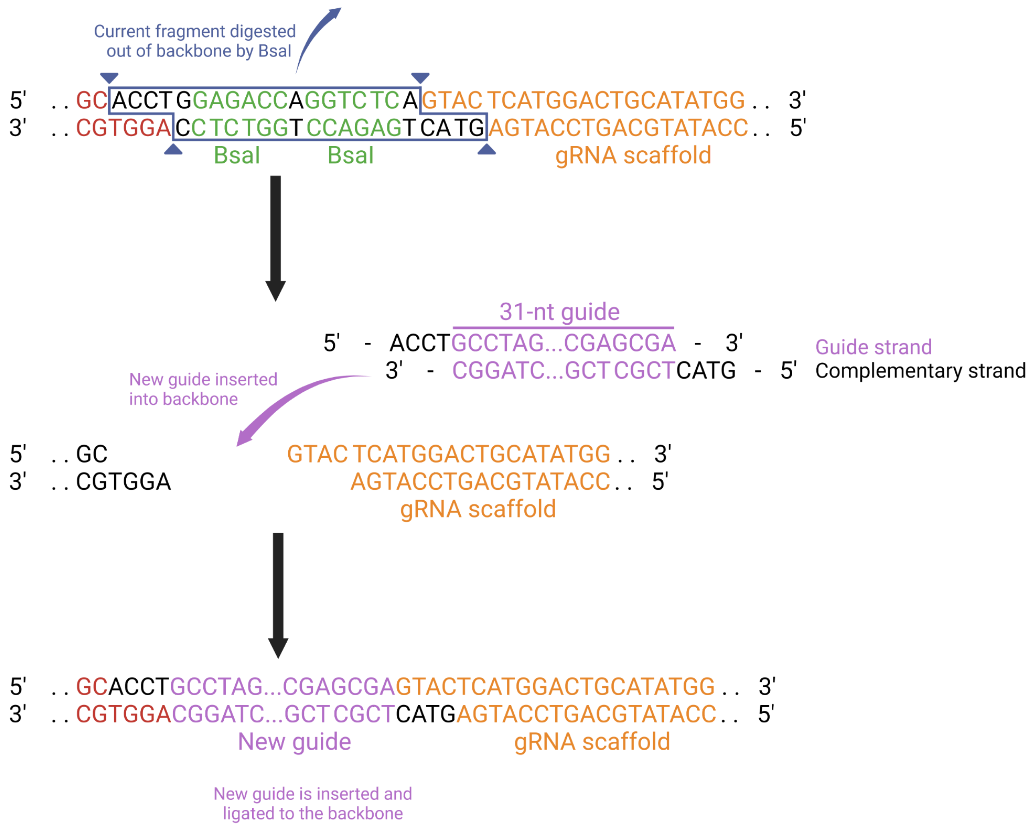



Figure 4: A forward and reverse primer ordered from IDT with substantial complementarity will bind to form the novel 31 bp spacer for the gRNA to be inserted. The overhangs of the gRNA represent the complementary sequence to the sticky ends created by restriction digestion of the backbone with BsaI at the BsaI cut sites. Through Golden Gate assembly, the new guide oligo can be inserted into the backbone and ligated, re-linearizing the vector with the new gRNA spacer sequence. This protocol is synonymous with that used in Ran et al. with BbsI cut sites [40].

There were several reasons for developing vectors with varying gRNAs. The primary purpose was to elucidate which gRNA spacer would most efficiently bind KRAS G12V; the spacer region of Cas7-11 is 31 bp in length. Hence, with a single nucleotide polymorphism (SNP) target such as KRAS G12V, there are hypothetically 31 completely complementary gRNAs that one could design. Further, partially complementary gRNAs could have similar if not superior binding efficiencies. Thus, developing a wide variety of vectors with different gRNAs was essential for determining the most efficient spacer for targeting the KRAS G12V transcript. A secondary consideration was determining the promiscuity of the Cas7-11 system by testing partially complementary gRNAs. If Craspase tolerated several bp mismatches between its gRNA and target, this would raise the possibility of off-target effects, heightening the potential safety risks of the system for *in vivo* applications. In addition to targeting gRNAs, we also developed scrambled gRNA vectors as controls to compare fusion protein cleavage efficiency between experimental vectors and the controls. Given the stochasticity of the sequence of the scrambled gRNA, the cleavage efficiency of these vectors was treated as the background cleavage level.

For targeting gRNAs, we first developed vectors where the KRAS G12V SNP is complementary to the +18 bp location on the spacer of the gRNA. We decided on the +18 location based on Strecker et al.'s experiments with two mismatches, where mismatches at the +17-18 window resulted in minimal cleavage of Csx30 [28] (Supplementary Materials, Figure S3 C). That is, the +17-18 bp location is highly sensitive, whereas locations closer to the scaffold RNA were much less so. Strecker et al. produced western blots demonstrating the sensitivity of the system with double mismatches at adjacent nucleotides. They tested with nucleotides (from 5' to 3') 1 and 2 to 21 and 22 mismatched in the gRNA. Mismatches after the +13 position saw maximum sensitivity from Craspase based on the minimal cleavage of Csx30 thereafter. Given these results, we chose to develop gRNAs where the KRAS G12V SNP is aligned with the +18 bp location on the gRNA.

All vectors were sequenced to confirm lack of mutagenesis and to confirm successful cloning.

Expression of Components in *S. Cerevisiae*

After sequence-verifying the transformation of all vectors in *E. coli*, the plasmids were recovered via DNA extraction, then transformed into the BY4741 *S. Cerevisiae* budding yeast strain. Expression of the vectors was modulated using both the auxotrophic selection markers and several yeast promoters encoded in the plasmids. The promoters and selection markers in each set of constructs are as follows: the KRAS constructs were regulated by a URA3 uracil auxotrophic selection marker and cyanamide-induced DDI2/3 promoter; the Csx29 construct was regulated by the HIS3 histidine auxotrophic selection marker and the galactose-induced GAL1/GAL10 bidirectional yeast promoter; the gasdermin-Csx30 fusion protein constructs were regulated by the HIS3 histidine auxotrophic selection marker and the GAL1/GAL10 bidirectional yeast promoter; and the Cas7-11 and gRNA were regulated by the LEU2 leucine auxotrophic selection marker and both the GAP and SNR52 constitutive promoters (GAP for Cas7-11 modulation and SNR52 for gRNA modulation).

The use of galactose-induced, cyanimide-induced, and constitutive promoters facilitated expression of specific vectors based on the selected induction. We manipulated five variables to test all combinations of the entire system; the manipulated variables included presence or absence of galactose, presence or absence of cyanimide, scrambled and several complementary gRNA sequences, presence of a gasdermin D or gasdermin E fusion protein vector, and presence of KRAS wild type vector or mutant. The presence of cyanimide was used to induce expression of the KRAS vector used in a given experiment; absence of cyanimide prevented transcription or translation of the KRAS gene. Galactose was used to induce expression of both GSDM fusion proteins and Csx29; absence of galactose prevented expression of GSDM fusions and Csx29. Insertion of a scrambled gRNA sequence in the pML107 backbone was used as a negative control to determine the extent of background cleavage with the Craspase system. Several complementary gRNA sequences were tested to determine the most efficient spacer for targeting KRAS G12V mRNA. Finally, we tested with both KRAS wild type and KRAs mutant G12V separately to determine the specificity of the Craspase system; if the gRNA-Cas7-11 were to induce pyroptosis with wild type KRAS, it would be necessary to have increased non-complementarity between the target RNA and the gRNA so as to minimize off-target binding as well as wild type KRAS attack. Testing every possible iteration of the variables would require 256 experiments to be run (32 per GSDM fusion) for each gRNA being tested. This was not feasible nor necessary in this case. The predicted outcome of each experiment is given in Table 3 of the Supplementary Information. After testing expression of each constituent separately (KRAS transcription, then Csx29, GSDM fusion protein, and Cas7-11 with gRNA expression), as well as testing for background cleavage with scrambled gRNA, we immediately tested the functionality of the whole system for each GSDM fusion protein. That is, we transformed *S. Cerevisiae* with all constituent vectors (KRAS G12V in pYes, the GSDM fusion of interest, Csx29 in pEsc, and Cas7-11 with complementary gRNA in pML107), then we induced with both galactose and cyanamide to express all elements of the system.

For these experiments, the gRNA was expected to hybridize with KRAS G12V mRNA, stimulating cleavage of the GSDM-Csx30 fusion protein, causing downstream pyroptosis of the cell. We performed protein extraction protocols targeting the 6xHis and FLAG tags at the termini of the protein constituents. This was done for all experiments to verify expression of each constituent as well as cleavage of the GSDM-Csx30 fusion proteins.

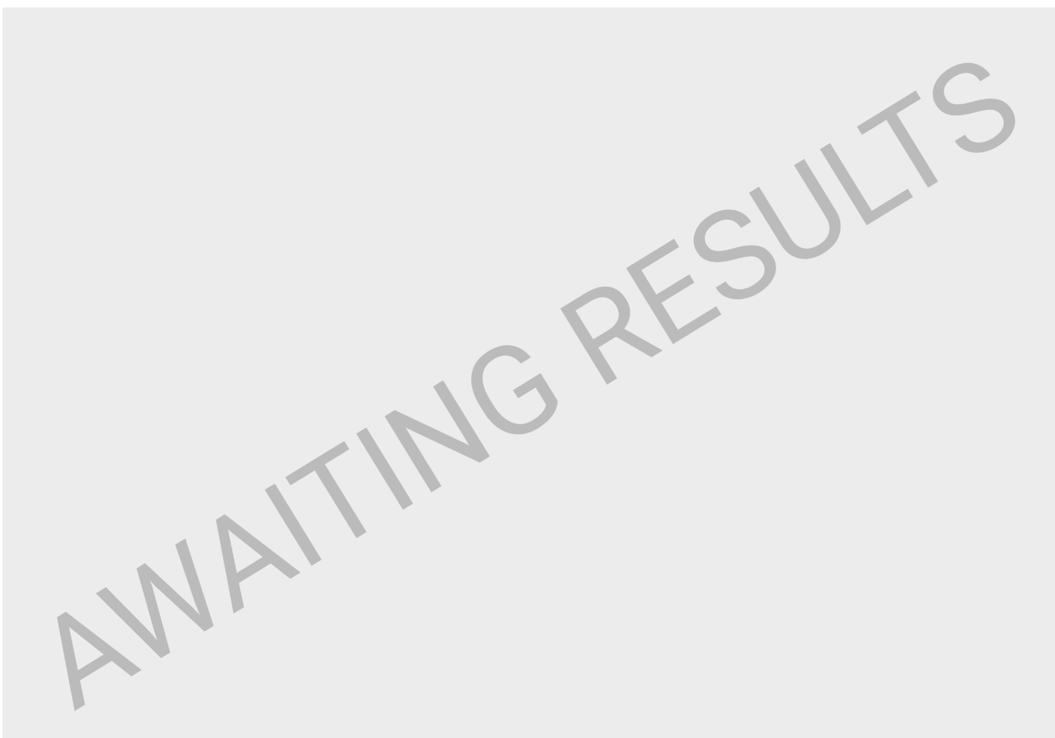
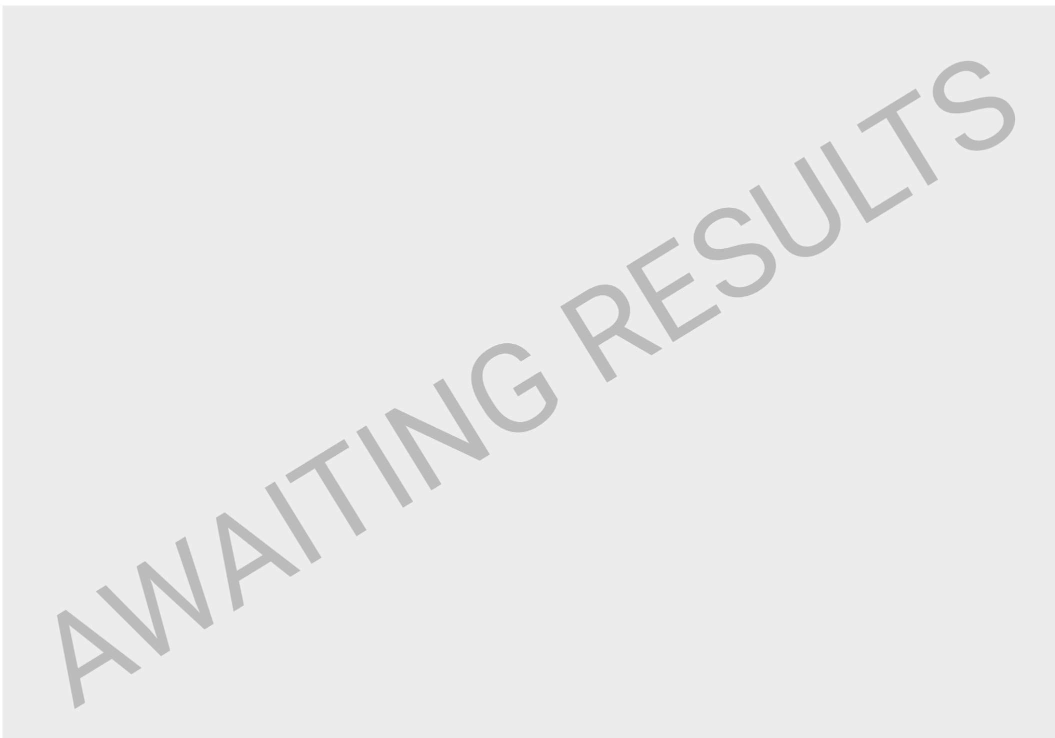



Figure 5: GSDM-Csx30 fusion protein cleavage in *S. Cerevisiae*. Negative control with scrambled gRNA instead of complementary gRNA. Positive controls with N-terminal domain GSDMD and C-terminal domain GSDME expressed in yeast (under galactose induction).

<u>Viability Assays: Propidium Iodide and SYTO 9 Staining for Growth Curves</u>

The cleavage efficiency for each GSDM fusion was determined using growth curves on triplicate samples. Cell viability was estimated every 30 minutes for 24 hours using propidium iodide and SYTO 9 staining. Higher cell viability translates to lower cleavage efficiency for the test GSDM fusion. In addition to testing cleavage efficiency, we tested binding efficiencies of different gRNAs to determine which would be optimal for inducing pyroptosis in the target. Our gRNA variant tests were also used to establish the inherent promiscuity of the Craspase system. Based on the results of growth curves, it would be possible to determine the optimal GSDM-Csx30 fusion protein and gRNA for high efficiency, low off-target effect pyroptosis in KRAS G12V-expressing *S. Cerevisiae*.

## Results & Discussion




<u>Planned Figures</u>

- Expression assays of Craspase components (western blots):
    - → + galactose - cyanamide: all components expressed, except KRAS/KRAS*
    - → + galactose + cyanamide: all components expressed
    - → - galactose - cyanamide: Only constitutive components expressed
    - → - galactose + cyanamide: constitutive components plus KRAS/KRAS*
- Cleavage of fusion proteins by craspase (western blot: distinct bands at the approximate size of the tagged terminus of each of the 8 fusion protein constructs):
    - → Control with scrambled gRNA (non-complementary, so no cleavage event)
        - » Should see one large band for each fusion (the uncut fusion)
    - → Experimental units with complementary gRNA targeting KRAS*
        - » Band at significantly smaller approximate size (cut fusion)

→ Experimental units with KRAS* gRNA , but cells expressing KRAS:
  » Should see one large band for each fusion (the uncut fusion)
  » If fusions are cut, the system is more promiscuous than expected; gRNAs must be re-engineered to prevent wild-type targeting

- Yeast growth curves (in triplicates):
  → Experimental units should see sharp decreases in viability over the testing period compared to controls with scrambled gRNA or KRAS in lieu of KRAS*

# Supplementary Information

**Table 1. Primers**

| ID No. | Name | Sequence (5' to 3') |
|---|---|---|
| 1 | GSDMD_CaspRem_mPCR_F | TTCCTGACAGcTGGGGTCCCT |
| 2 | GSDMD_CaspRem_mPCR_R | GTTGTGGAGGCACCTCATC |
| 3 | GSDME_CaspRem_mPCR_F | GACATGCCAGcTGCTGCGCAT |
| 4 | GSDME_CaspRem_mPCR_R | TATGAATGCAAACTCTCGAAAGACCAG |
| 5 | GSDMD_MCS_GSD_GB_F | AAAAAAATGGACTACAAGGATGACGACGACAAAatggggtcggcctttgagcg |
| 6 | GSDMD_MCS_GSD_GB_R | tgcggccctctagatgcatgctagtggggctcctggctcag |
| 7 | GSDMD_MCS_pYes_GB_F | tgagccaggagccccactagcatgcatctagagggccgc |
| 8 | GSDMD_MCS_pYes_GB_R | TCGTCGTCATCCTTGTAGTCCATTTTTTTCGAAACTAAGTTCTTGGTGTTTTAAAAC |
| 9 | GSDME_MCS_GSD_GB_F | GAACTTAGTTTCGAAAAAAAatgtttgccaaagcaaccag |
| 10 | GSDME_MCS_GSD_GB_R | TCATTTGTCGTCGTCATCCTTGTAGTCtgaatgttctctgcctaaagcac |
| 11 | GSDME_MCS_pYes_GB_F | GACTACAAGGATGACGACGACAAATGAcatgcatctagagggcc |
| 12 | GSDME_MCS_pYes_GB_R | gctttggcaaacatTTTTTTTCGAAACTAAGTTCTTGGTGTTTTAAAAC |
| 13 | Csx30_MCS_Csx30_GB_F | AAAAAAATGGACTACAAGGATGACGACGACAAAatgAATACCACAACATATAACACTACGACTGAT |
| 14 | Csx30_MCS_Csx30_GB_R | tgcggccctctagatgcatgTTAGCCTTCTTGCGGAATGG |
| 15 | Csx30_MCS_pYes_GB_F | CCATTCCGCAAGAAGGCTAAcatgcatctagagggccgca |

| | | |
|---|---|---|
| 16 | Csx30_MCS_pYes_GB_R | TTTGTCGTCGTCATCCTTGTAGTCCATTTTTTTTCGAAACTAAGTTCTTGGTG TTTTAAAACTAAA |
| 17 | GDDX30_C-term_C30_F | ctgcggaggggGGTTCTTCTACAGACGACGTTAAGCCGC |
| 18 | GDDX30_C-term_C30_R | tcagtgaacgcAGAAGAACCGCCTTCTTGCGGAATGGTGAC |
| 19 | GDDX30_C-term_GSD_F | CGCAAGAAGGCGGTTCTTCTgcgttcactgaagacttccag |
| 20 | GDDX30_C-term_GSD_R | ACGTCGTCTGTAGAAGAACCcccctccgcagggac |
| 21 | GDDx30_N-term_C30_F | tccagccacccGGTTCTTCTACAGACGACGTTAAGCCGC |
| 22 | GDDx30_N-term_C30_R | tggcctgtcgcAGAAGAACCGCCTTCTTGCGGAATGGTG |
| 23 | GDDx30_N-term_GSD_F | CGCAAGAAGGCGGTTCTTCTgcgacaggccacaagc |
| 24 | GDDx30_N-term_GSD_R | ACGTCGTCTGTAGAAGAACCgggtggctggaaggtcc |
| 25 | GDDX30_30rep_C30_F | agcgttccacgGGTTCTTCTACAGACGACGTTAAGCCG |
| 26 | GDDX30_30rep_C30_R | aggcacctcatAGAAGAACCGCCTTCTTGCGGAATGG |
| 27 | GDDX30_30rep_GSD_F | CGCAAGAAGGCGGTTCTTCTatgaggtgcctccacaac |
| 28 | GDDX30_30rep_GSD_R | ACGTCGTCTGTAGAAGAACCcgtggaacgcttgtgg |
| 29 | GDDX30_CaspR_C30_F | gcctccacaacGGTTCTTCTACAGACGACGTTAAGCCG |
| 30 | GDDX30_CaspR_C30_R | tccgcagggacAGAAGAACCGCCTTCTTGCGGAATGGTG |
| 31 | GDDX30_CaspR_GSD_F | CGCAAGAAGGCGGTTCTTCTgtccctgcggaggg |
| 32 | GDDX30_CaspR_GSD_R | ACGTCGTCTGTAGAAGAACCgttgtggaggcacctcatc |
| 33 | GDEX30_Cterm_C30_F | caggatggttcttctACAGACGACGTTAAGCCGC |
| 34 | GDEX30_Cterm_C30_R | tggtccagaagaaccGCCTTCTTGCGGAATGGTG |
| 35 | GDEX30_Cterm_GSD_F | GAAGGCggttcttctggaccattaagtgttttaaagcaagc |
| 36 | GDEX30_Cterm_GSD_R | GTCTGTagaagaaccatcctgggaagatatcccatgc |
| 37 | GDEx30_NTerm_GSE_F | GAAGGCggttcttctattgactctgtctacctggacc |
| 38 | GDEx30_NTerm_GSE_R | GTCTGTagaagaacctctcttcttgttctcgaagccac |

| | | |
|---|---|---|
| 39 | GDEx30_NTerm_C30_F | aagagaggttcttctACAGACGACGTTAAGCCGC |
| 40 | GDEx30_NTerm_C30_R | gtcaatagaagaaccGCCTTCTTGCGGAATGGTGAC |
| 41 | GDEX30_30rep_C30_F | gtctacggttcttctACAGACGACGTTAAGCCGC |
| 42 | GDEX30_30rep_C30_R | tatgaaagaagaaccGCCTTCTTGCGGAATGGTGAC |
| 43 | GDEX30_30rep_GSE_F | GAAGGCggttcttctttcatagacatgccagctgctg |
| 44 | GDEX30_30rep_GSE_R | GTCTGTagaagaaccgtagacagagtcaattctcttcttgttctc |
| 45 | GDEx30_CaspR_C30_F | ttcataggttcttctACAGACGACGTTAAGCCGC |
| 46 | GDEx30_CaspR_C30_R | atgcgcagaagaaccGCCTTCTTGCGGAATGGTGAC |
| 47 | GDEx30_CaspR_GSE_F | GAAGGCggttcttctgcgcatgggatatcttccc |
| 48 | GDEx30_CaspR_GSE_R | GTCTGTagaagaacctatgaatgcaaactctcgaaagacc |
| 49 | GSDMD_Cterm_rem_F | TAGcatgcatctagagggcc |
| 50 | GSDMD_Cterm_rem_R | agctgtcaggaagttgtggagg |
| 51 | GSDME_Cterm_rem_F | GACTACAAGGATGACGACGAC |
| 52 | GSDME_Cterm_rem_R | AGCTGGCATGTCTATGAATG |
| 53 | pYES_seq_F | ACCAGTTCCCTGAAATTATTCCCC |
| 54 | pYES_seq_R | gggacctagacttcaggttgtctaac |
| 55 | Csx29_Nterm_mPCR_F | gccgaattcaaaaaaatgcatcaccatcaccatcacatgagcaatcctatcagagatatccaggac |
| 56 | Csx29_CTerm_mPCR_R | atagcggccgctcattcggcaggaaagccaat |
| 57 | Csx29_pEsc_Seq_F | agacaagccgacaaccttgattgg |
| 58 | Csx29_pEsc_Seq_R | ggggctctttacatttccacaac |
| 59 | KRAS_Nterm_mPCR_F | atacccgggaaaaaaatgactgaatataaacttgtggtagttgg |
| 60 | KRAS_Cterm_mPCR_R | atagtcgacttagtgatggtgatggtgatgcataattacacactttgtctttgacttctt |
| 61 | KRAS_D12G_mPCR_F | gttggagctggtggcgtaggc |

| | | |
|---|---|---|
| 62 | KRAS_D12G_mPCR_R | taccacaagtttatattcagtcatggtgc |
| 63 | KRAS_pEsc_SEQ_F | aaccactttaactaatactttcaacattttcgg |
| 64 | KRAS_pEsc_SEQ_R | gacttcaggttgtctaactccttcc |
| 65 | GB_gRNA_F | cgtggacaccgtgggcataaactctcgaggcgaatttctt |
| 66 | GB_gRNA_R | gcagtgaaagataaatgatcttgatgtcacggaaccgaga |
| 67 | GB_pML107_F | tctcggttccgtgacatcaagatcatttatctttcactgcgg |
| 68 | GB_pML107_R | GTGATGGTGATGGTGATGCATTTTTTTgagatctaattcgatccatatcagatc |
| 69 | GB_DisCas7-11_F | AAATGCATCACCATCACCATCACatgacgactactatgaagatttcaattgaattcc |
| 70 | GB_DisCas7-11_R | aagaaattcgcctcgagagtttatgcccacggtgtccacg |
| 71 | Cas_pML_GAP_seq_F | gctgaaaaaaaaggttgaaaccagttccc |
| 72 | Cas_pML_AHD1_seq_R | cctgacctacaggaaagagttactcaag |
| 73 | GSDME_Cutsite_Seq_F | acggaggatgggaatgtcaccaa |
| 74 | KRAS_pYes_SynthInsert_F | atgtgtataactcgagcatgcatctagagg |
| 75 | KRAS_pYes_SynthInsert_R | TTTAACCTTTGAAgtattgtagaatctttattgttcggagcag |

| **Table 2. Gene amino acid sequences** | |
|---|---|
| Human Gasdermin D (GSDMD) | MGSAFERVVRRVVQELDHGGEFIPVTSLQSSTGFQPYCLVVRKPSSSW FWKPRYKCVNLSIKDILEPDAAEPDVQRGRSFHFYDAMDGQIQGSVEL AAPGQAKIAGGAAVSDSSSTSMNVYSLSVDPNTWQTLLHERHLRQPE HKVLQQLRSRGDNVYVVTEVLQTQKEVEVTRTHKREGSGRFSLPGAT CLQGEGQGHLSQKKTVTIPSGSTLAFRVAQLVIDSDLDVLLFPDKKQRT FQPPATGHKRSTSEGAWPQLPSGLSMMRCLHNFLTDGVPAEGAFTED FQGLRAEVETISKELELLDRELCQLLLEGLEGVLRDQLALRALEEALEQ GQSLGPVEPLDGPAGAVLECLVLSSGMLVPELAIPVVYLLGALTMLSET QHKLLAEALESQTLLGPLELVGSLLEQSAPWQERSTMSLPPGLLGNSW GEGAPAWVLLDECGLELGEDTPHVCWEPQAQGRMCALYASLALLSGL SQEPH |
| Human Gasdermin E (GSDME) | MFAKATRNFLREVDADGDLIAVSNLNDSDKLQLLSLVTKKKRFWCWQ RPKYQFLSLTLGDVLIEDQFPSPVVVESDFVKYEGKFANHVSGTLETAL GKVKLNLGGSSRVESQSSFGTLRKQEVDLQQLIRDSAERTINLRNPVLQ QVLEGRNEVLCVLTQKITTMQKCVISEHMQVEEKCGGIVGIQTKTVQVS ATEDGNVTKDSNVVLEIPAATTIAYGVIELYVKLDGQFEFCLLRGKQGG FENKKRIDSVYLDPLVFREFAFIDMPDAAHGISSQDGPLSVLKQATLLL ERNFHPFAELPEPQQTALSDIFQAVLFDDELLMVLEPVCDDLVSGLSPT VAVLGELKPRQQQDLVAFLQLVGCSLQGGCPGPEDAGSKQLFMTAYFLV SALAEMPDSAAALLGTCCKLQIIPTLCHLLRALSDDGVSDLEDPTLTPL KDTERFGIVQRLFASADISLERLKSSVKAVILKDSKVFPLLLCITLNGLC |

| | |
|---|---|
| | ALGREHS |
| Csx30 from *Desulfonema ishimotonii* | MNTTTYNTTTDALLEWGKVYFQKEDFSEFLDNLEAYISDAGDSLKDE<br>LESGVEKLVLGIKSAEAVIFGEAVIGTTPENEAWYDAEESFLTLDCAVW<br>LSQALDRVVRRQDASLADSLIARLDEAINRVAEKLYADNLSPLRFSSLN<br>EIRRSALEATDEKYHYLFPWHGAACDVDENILLILTEEYHLIGADKAGA<br>NLSEELRGDLPFIFAELERDEVLRAYVEKENALSLALENTMREHWAFG<br>LLEAARDEGYNHPYPADVGMRIHQVARAVFSQTNLSPAERLAVAIAGA<br>CFTPEISEDRRLEILLDCEERVCEIEAPTGDDTSVRVIKDLKALADHRV<br>RHEIPAESLVSLWFEQIEAAGTDFDTKTPMDELVLRMLSDNVITLSVD<br>RKAASQTETDDVKPQKGKIIPFPVPDIANDEVEYQKAVGMKKDKKAA<br>NDSKVKFPGLLEIQGCRDGDKAILLEDTDDAAANHRKLFSILKAGKLN<br>SAFFIQSDDGEWVESESKPTMEDNRIILHDSHHSSFVWILDTGSMQLR<br>QSVKCVKDALNKKTGSAKKLKPKTMIVWVTIPQEG |
| Csx29 (TPR-CHAT) | MSNPIRDIQDRLKTAKFDNKDDMMNLASSLYKYEKQLMDSSEATLCQ<br>QGLSNRPNSFSQLSQFRDSDIQSKAGGQTGKFWQNEYEACKNFQTHK<br>ERRETLEQIIRFLQNGAEEKDADDLLLKTLARAYFHRGLLYRPKGFSVP<br>ARKVEAMKKAIAYCEIILDKNEEESEALRIWLYAAMELRRCGEEYPENF<br>AEKLFYLANDGFISELYDIRLFLEYTEREEDNNFLDMILQENQDRERLF<br>ELCLYKARACFHLNQLNDVRIYGESAIDNAPGAFADPFWDELVEFIRM<br>LRNKKSELWKEIAIKAWDKCREKEMKVGNNIYLSWYWARQRELYDL<br>AFMAQDGIEKKTRIADSLKSRTTLRIQELNELRKDAHRKQNRRLEDKL<br>DRIIEQENEARDGAYLRRNPPCFTGGKREEIPFARLPQNWIAVHFYLN<br>ELESHEGGKGGHALIYDPQKAEKDQWQDKSFDYKELHRKFLEWQEN<br>YILNEEGSADFLVTLCREIEKAMPFLFKSEVIPEDRPVLWIPHGFLHRL<br>PLHAAMKSGNNSNIEIFWERHASRYLPAWHLFDPAPYSREESSTLLKN<br>FEEYDFQNLENGEIEVYAPSSPKKVKEAIRENPAILLLLCHGEADMTNP<br>FRSCLKLKNKDMTIFDLLTVEDVRLSGSRILLGACESDMVPPLEFSVDE<br>HLSVSGAFLSHKAGEIVAGLWTVDSEKVDECYSYLVEEKDFLRNLQEW<br>QMAETENFRSENDSSLFYKIAPFRIIGFPAE |
| dCas7-11 | MTTTMKISIEFLEPFRMTKWQESTRRNKNNKEFVRGQAFARWHRNK<br>KDNTKGRPYITGTLLRSAVIRSAENLLTLSDGKISEKTCCPGKFDTEDK<br>DRLLQLRQRSTLRWTDKNPCPDNAETYCPFCELLGRSGNDGKKAEKK<br>DWRFRIHFGNLSLPGKPDFDGPKAIGSQRVLNRVDFKSGKAHDFFKAY<br>EVDHTRFPRFEGEITIDNKVSAEARKLLCDSLKFTDRLCGALCVIRFDE<br>YTPAADSGKQTENVQAEPNANLAEKTAEQIISILDDNKKTEYTRLLAD<br>AIRSLRRSSKLVAGLPKDHDGKDDHYLWDIGKKKKDENSVTIRQILTTS<br>ADTKELKNAGKWREFCEKLGEALYLKSKDMSGGLKITRRILGDAEFHG<br>KPDRLEKSRSVSIGSVLKETVVCGELVAKTPFFFGAIDEDAKQTALQVLL<br>TPDNKYRLPRSAVRGILRRDLQTYFDSPCNAELGGRPCMCKTCRIMRG<br>ITVMDARSEYNAPPEIRHRTRINPFTGTVAEGALFNMEVAPEGIVFPFQ<br>LRYRGSEDGLPDALKTVLKWWAEGQAFMSGAASTGKGRFRMENAKY<br>ETLDLSDENQRNDYLKNWGWRDEKGLEELKKRLNSGLPEPGNYRDP<br>KWHEINVSIEMASPFINGDPIRAAVDKRGTAVVTFVKYKAEGEEAKPV<br>CAYKAESFRGVIRSAVARIHMEDGVPLTELTHSDCECLLCQIFGSEYEAG<br>KIRFEDLVFESDPEPVTFDHVAIDRFTGGAADKKKFDDSPLPGSPARPL<br>MLKGSFWIRRDVLEDEEYCKALGKALADVNNGLYPLGGKSAIGYGQVK<br>SLGIKGDDKRISRLMNPAFDETDVAVPEKPKTDAEVRIEAEKVYYYPHY<br>FVEPHKKVEREEKPCGHQKFHEGRLTGKIRCKLITKTPLIVPDTSNDD<br>FFRPADKEARKEKDEYHKSYAFFRLHKQIMIPGSELRGMVSSVYETVT<br>NSCFRIFDETKRLSWRMDADHQNVLQDFLPGRVTADGKHIQKFSETA<br>RVPFYDKTQKHFDILDEQEIAGEKPVRMWVKRFIKRLSLVDPAKHPQ<br>KKQDNKWKRRKEGIATFIEQKNGSYYFNVVTNNGCTSFHLWHKPDN<br>FDQEKLEGIQNGEKLDCWVRDSRYQKAFQEIPENDPDGWECKEGYLH<br>VVGPSKVEFSDKKGDVINNFQGTLPSVPNDWKTIRTNDFKNRKRKNE<br>PVFCCEDDKGNYYTMAKYCETFFFDLKENEEYEIPEKARIKYKELLRV<br>YNNNPQAVPESVFQSRVARENVEKLKSGDLVYFKHNEKYVEDIVPVRI<br>SRTVDDRMIGKRMSADLRPCHGDWVEDGDLSALNAYPEKRLLLRHP<br>KGLCPACRLFGTGSYKGRVRFGFASLENDPEWLIPGKNPGDPFHGGPV<br>MLSLLERPRPTWSIPGSDNKFKVPGRKFYVHHHAWKTIKDGNHPTT<br>GKAIEQSPNNRTVEALAGGNSFSFEIAFENLKEWELGLLIHSLQLEKGL<br>AHKLGMAKSMGFGSVEIDVESVRLRKDWKQWRNGNSEIPNWLGKGF<br>AKLKEWFRDELDFIENLKKLLWFPEGDQAPRVCYPMLRKKDDPNGN<br>SGYEELKDGEFKKEDRQKKLTTPWTPWA* |

# Table 3: Expected Results for Each Variable Change

| | Status | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Gal+/- | + | - | + | - | + | - | + | - |
| Cyan+/- | + | + | - | - | + | + | - | - |
| gRNA Type | Complementary | Complementary | Complementary | Complementary | Scrambled | Scrambled | Scrambled | Scrambled |
| Presense of GSDM Fusion Vector | + | + | + | + | + | + | + | + |
| KRAS Variant | G12D | G12D | G12D | G12D | G12D | G12D | G12D | G12D |
| Results ↓ | | | | | | | | |
| KRAS Variant Transcription | + | + | - | - | + | + | - | - |
| Cas7-11+gRNA Expression | + | + | + | + | + | + | + | + |
| Csx29 Expression | + | - | + | - | + | - | + | - |
| GSDM Fusion Expression | + | - | + | - | + | - | + | - |
| gRNA-Target Binding | + | + | + | - | - | - | - | - |
| GSDM Fusion Cleavage | + | - | - | - | - | - | - | - |
| Cell Death | + | - | - | - | - | - | - | - |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Gal+/- | + | - | + | - | + | - | + | - |
| Cyan+/- | + | + | - | - | + | + | - | - |
| gRNA Type | Complementary | Complementary | Complementary | Complementary | Scrambled | Scrambled | Scrambled | Scrambled |
| Presense of GSDM Fusion Vector | - | - | - | - | - | - | - | - |
| KRAS Variant | G12D | G12D | G12D | G12D | G12D | G12D | G12D | G12D |
| Results ↓ | | | | | | | | |
| KRAS Variant Transcription | + | + | - | - | + | + | - | - |
| Cas7-11+gRNA Expression | + | + | + | + | + | + | + | + |
| Csx29 Expression | + | - | + | - | + | - | + | - |
| GSDM Fusion Expression | + | - | + | - | + | - | + | - |
| gRNA-Target Binding | + | + | - | - | - | - | - | - |
| GSDM Fusion Cleavage | + | - | - | - | - | - | - | - |
| Cell Death | + | - | - | - | - | - | - | - |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Gal+/- | + | - | + | - | + | - | + | - |
| Cyan+/- | + | + | - | - | + | + | - | - |
| gRNA Type | Complementary | Complementary | Complementary | Complementary | Scrambled | Scrambled | Scrambled | Scrambled |
| Presense of GSDM Fusion Vector | + | + | + | + | + | + | + | + |
| KRAS Variant | wt | wt | wt | wt | wt | wt | wt | wt |
| Results ↓ | | | | | | | | |
| KRAS Variant Transcription | + | + | - | - | + | + | - | - |
| Cas7-11+gRNA Expression | + | + | + | + | + | + | + | + |
| Csx29 Expression | + | - | + | - | + | - | + | - |
| GSDM Fusion Expression | + | - | + | - | + | - | + | - |
| gRNA-Target Binding | + | + | - | - | - | - | - | - |
| GSDM Fusion Cleavage | + | - | - | - | - | - | - | - |
| Cell Death | + | - | - | - | - | - | - | - |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Gal+/- | + | - | + | - | + | - | + | - |
| Cyan+/- | + | + | - | - | + | + | - | - |
| gRNA Type | Complementary | Complementary | Complementary | Complementary | Scrambled | Scrambled | Scrambled | Scrambled |
| Presense of GSDM Fusion Vector | - | - | - | - | - | - | - | - |
| KRAS Variant | wt | wt | wt | wt | wt | wt | wt | wt |
| Results ↓ | | | | | | | | |
| KRAS Variant Transcription | + | + | - | - | + | + | - | - |
| Cas7-11+gRNA Expression | + | + | + | + | + | + | + | + |
| Csx29 Expression | + | - | + | - | + | - | + | - |
| GSDM Fusion Expression | + | - | + | - | + | - | + | - |
| gRNA-Target Binding | + | + | - | - | - | - | - | - |
| GSDM Fusion Cleavage | + | - | - | - | - | - | - | - |
| Cell Death | + | - | - | - | - | - | - | - |

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